Company Background:

AutoHDR is an AI real estate photo editor used by **professional real estate photographers.**

We train custom models that take in RAW, HDR photos and output high-res photos in the classic real estate style. We also automatically remove cameras, replace the sky, add fire in the fireplace - and conduct the long tail of real estate specific edits.

20% of US real estate photos are currently edited by AutoHDR.

Background

Nearly all photography companies shoot photos and videos for their real estate agent clients.

A big pain point is that video shooting and editing is HARD - and when they do learn to do it - it takes years of training on shooting and editing techniques. When the photographers go to grow their business, they must then train novice videographers how to use a gimbal, shoot complex movements, and find/train new editors who understand creative vision and editing.

Even with months and years of experience and training - most photography companies still don't achieve the level of excellence from the top videographers in the space (click to see a sample of the pinnacle of work).

They look up to the top videographers who have the best creative vision and years of experience. They will pay thousands of dollars to buy their courses and fly them out to train their team. Even then - it's time consuming, expensive, and rarely are the novice photographers able to re-create the creative excellence of these top creative influencers in the space. Everyone strives to achieve their level of quality - but they lack the creative vision.

The Hackathon Track

AutoHDR's New Product: Automated Photo-to-Video

We'd like participants to build automated photo-to-video applications. We are primarily looking for strong systems engineering, creative problem solving, and consistent professional level output.

Below is a detailed description of the product we are working on. You **do not** need to build this exact product, as long as a photo-to-video generation pipeline is core to your submissions value, you will qualify for the track. However, we include this to give you a better idea of what types of builds are interesting for us and give you a starting point to work from.
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We are very close with the biggest real estate videographers/editors in the world and aim to build a system that:

1). Allows them to create videos from sample photoshoots using the video models, prompts, edits, song, and other creative elements that they want, to create a "style" template. We can build the architecture/infrastructure as engineers, but the creatives need to pick the best prompts, models, sequences, songs, etc to give the viewer goosebumps.

2). Allows end users in the future to select from a *library of styles* when uploading their photoshoots for automated video editing.

End product vision:

1. A user uploads their edited property photos
2. The user selects the ***video style*** from their favorite creative videographer/influencer
3. They get back a fully edited video, generated from their photos, that *does not look AI generated* and perfectly adheres to a video style that was created by their favorite videographer.

For example, one style could be this video as "*Ski House Style* | By JT Visuals.

When selected for editing - the user's photoshoot would get all the same movements, on the same angles, cut up to the same song with all the same effects.

Think about it like this: you upload your photos, and it's like the person who built the style template has shot and edited the video for you.

The video prompts are the shots.

The clips they're applied to, the order of the clips, and the cuts to the song are the edits.

What this will require:

1. We need a way for real estate videographers to create an edit *style*. This is where they cook creatively and create their style.
    a. **For example:** give them a demo photoshoot, let them turn each clip into a video, edit it, and capture all their creative decisions for reproducibility
    b. **Another example:** let them upload a past video and we recreate that style
    c. **Another example:** let them create an outline of the storyboard at every timestamp that then gets recreated.
        i. These examples are *not* meant to bias you towards any approach - rather help you understand the goal: reproduce their exact video they spent hours pouring creative energy into.

2. We need a way to capture their decisions and EXACTLY reproduce the style they created. Down to the song, the movement of the shots, and any other creative decisions. If you play their original edit and future, generated edits side by side - they should look nearly identical - just on a different house
3. A way of then re-producing the edits and creative decisions on future photoshoots, which then product the video edit

Final things to note:

- When we have done this with creatives, they want to be able to take a mediocre photo, and then manipulate it into looking like how they would do it. This could mean:
    - their composition
    - their lens
    - the angle they'd get
    - the exact movement they'd use
    - the speed of the movement
    - shot with the lighting for the time of day they want

We observed they would do something like:

1. Take a normal edited photo from our software (if they just used this, it would make it obvious it's an AI video. The **goal** is the outputs are indistinguishable from a pro videographer.)

Input image:



2. Prompt Nano Banana (Google's image model)

Prompt + Output





3. Taking result of the last edit, then prompt Nano Banana again

Prompt + Output

Extreme close-up detail shot with smooth tracking camera following harsh directional light as it grows and spreads across textured surface. Crisp shadow edges crawl and shift in real time. Camera moves with the light's path revealing texture in wood grain, fabric weave, architectural detail. Editorial film style. Neutral white balance, balanced exposure — deep dramatic shadows without underexposure. Shallow depth of field.



4. Taking the final photo, then generating the video

Prompt: “Very slow truck right, time-lapse light progression, camera slides laterally while light shifts across the space, shadows gradually move and lengthen, parallax effect, consistent exposure, stable motion, cinematic, photorealistic”

Video Output

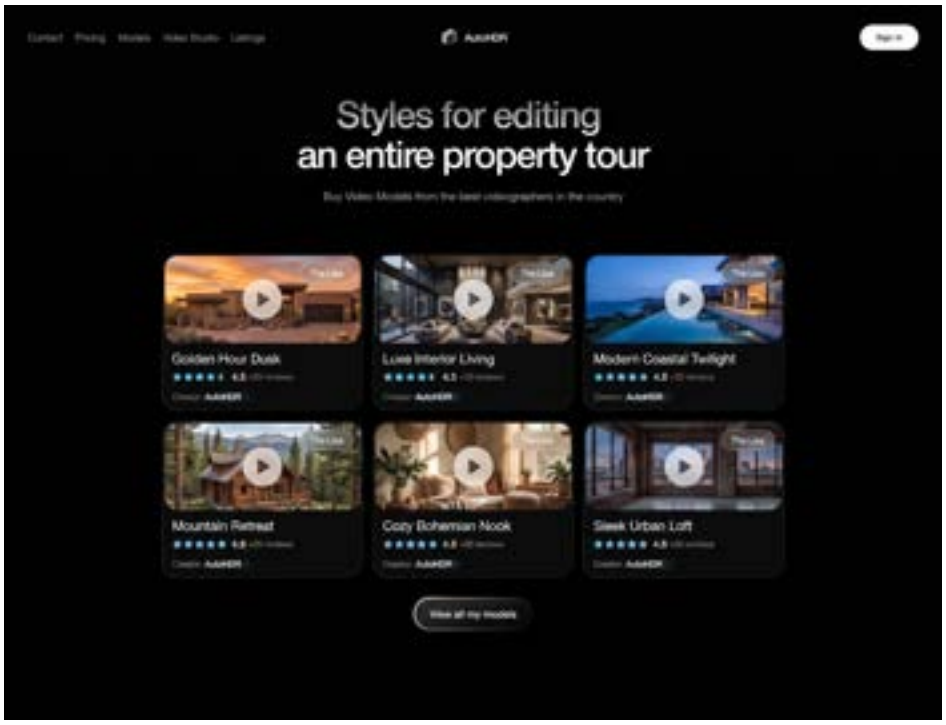
Watch here: https://youtu.be/M1bcFs3SjjU

Just in that sequence - we took an entry level photographer’s shot and made a cinematic timelapse detail shot that would take years of training, thousands of dollars in gear, and optimal lighting conditions to capture in real life.

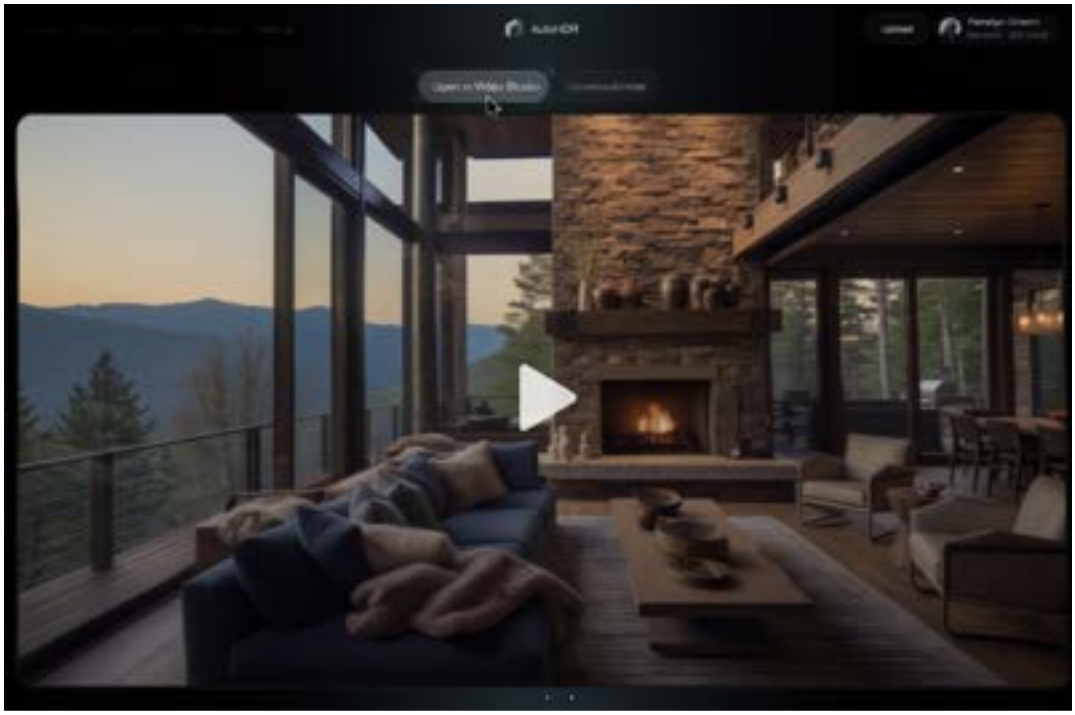
To help you **further visualize how this product will look/feel** - here are our UI mockups for this feature, and how we picture the overall experience once live for our users.

Once you understand this - your job is just to achieve this outcome - and figure out the zero -> one.

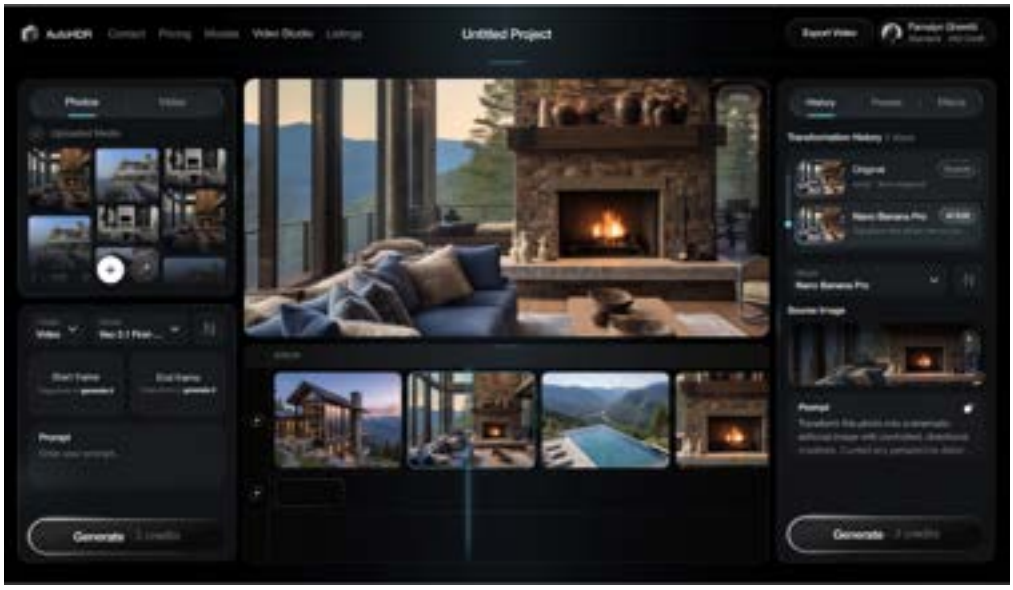
1. The user selects their style from their favorite creator, or even their own saved video style



2. The user gets the full video back - done - fully automated.



3. They can open the video in a web edit (if they want). If they want to make any changes - swap the room, change a shot, remove a room - the entire video project file will be in a web editor they can click into. This is *optional* - but gives the full granularity to make edits on the video, and re-export. We want a fully automated solution, but agents & homeowners often have subjective revision requests; this gives flexibility for those.



Resources:

Folder of test photoshoots, randomly sampled across users:

To get 12 test photoshoots, download them from dropbox:

https://www.dropbox.com/scl/fo/t09fmh711k8tde9bgjvko/ABZ0KRMaye6FV3WMEWn5QI0?rlkey=bw4ru1fr6lh8nyaeq9x38pra5&st=08mvq5qn&dl=0

Prompt starter kit from time we spent with the top 2 creatives iterating on videos:

VIDEO MOVEMENTS

(photo -> video models)

Dolly In (Path-Based)

```
Super smooth camera moves forward in straight line through space, cinematic
```

Dolly In (timelapse)

```
Super smooth camera moves forward in straight line through space,time-lapse light progression, shadows gradually move and lengthen, parallax effect, consistent exposure, stable motion, cinematic, photorealistic
```

Drone Top Down Descend

```
Super smooth camera descends in a straight line through space, cinematic
```

Timelapse Variant

```
Super smooth camera moves forward in straight line through space, cinematic. As time progresses, the sun lowers as the day progresses as a timelapse.
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Dolly Out (Path-Based)

```
Super smooth camera moves backward in straight line revealing space, cinematic
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Slider Truck Left to Right (Path-Based)

```
Super smooth camera glides horizontally from left to right, parallel path, cinematic
```

Slider Truck Right to Left (Path-Based)

```
Super smooth camera glides horizontally from right to left, parallel path, cinematic
```

Parallax Orbit (Path-Based)

```
Super smooth camera travels in arc around subject, subject stays centered, cinematic
```

Dolly In + Pan Right (Path-Based)

```
Super smooth camera moves forward while rotating gradually right, curved path, cinematic
```

Crane Up (Path-Based)

```
Super smooth camera rises vertically upward, straight vertical path, cinematic
```

Crane Down (Path-Based)

```
Super smooth camera descends vertically downward, straight vertical path, cinematic
```

Orbit (Hyperlapse)

```
Super smooth camera travels in arc around subject, subject stays centered, sky hyperlapses naturally in the background, cinematic
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Tuned for Kling O1:
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Slider Prompt:

Very slow truck right, time-lapse light progression, camera slides laterally while light shifts across the space, shadows gradually move and lengthen, parallax effect, consistent exposure, stable motion, cinematic, photorealistic

Wide slide

Wide interior shot with slow trucking movement side to side as harsh directional light moves and expands across modern living space. Dramatic shadows crawl and shift across curved sofa, wood paneling, walls, and furnishings. Crisp shadow edges in motion. Camera tracks laterally through scene while light travels. Editorial film style. Neutral white balance, balanced exposure — deep shadows without underexposure. Smooth parallel motion. Atmospheric architectural cinematography.

Wide dolly in

Wide interior shot with a slow and smooth dolly in. Dramatic shadows crawl and shift across curved sofa, wood paneling, walls, and furnishings. Crisp shadow edges in motion. Camera tracks laterally through scene while light travels. Editorial film style. Neutral white balance, balanced exposure — deep shadows without underexposure. Smooth parallel motion. Atmospheric architectural cinematography.

Tight truck

Tight interior shot with slow trucking movement side to side as harsh directional light moves and expands across modern living space. Dramatic shadows crawl and shift across curved sofa, wood paneling, walls, and furnishings. Crisp shadow edges in motion. Camera tracks laterally through scene while light travels. Editorial film style. Neutral white balance, balanced exposure — deep shadows without underexposure. Smooth parallel motion. Atmospheric architectural cinematography.

CINEMATIC PHOTO PROMPTS

(Image to image models)

```
Transform this photo into a cinematic editorial image with harsh, directional shadows and crisp light shaping. Correct any perspective distortion—ensure vertical lines are truly vertical and horizontal lines are level. Maintain the exact scene composition after correction and preserve the original white balance. Balance overall exposure with
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intention: raise interior midtones subtly for improved readability and presence, while preserving deep, sculpted shadow structure and strong contrast. The interior should feel brighter and more intentional, not flat or evenly lit—shadows must remain graphic, directional, and editorial. Highlights should stay controlled and natural.Apply intentional, filmic window pulls that reveal deep, rich exterior views—preserve sky density, environmental color, and contrast beyond the glass. Exterior scenes should feel dimensional and weighty, never washed out or pastel. Window highlights must roll off smoothly with realistic falloff; avoid haloing, edge glow, or global tonal compression. Do not flatten contrast or lift blacks globally. Window recovery should feel localized, natural, and optically believable—similar to a well-exposed negative rather than HDR processing. Preserve all architectural and interior details. The scene should feel well-lit yet moody, polished and cinematic. Derive all lighting direction strictly from visible sources in the frame—windows, doors, architectural openings, and practical fixtures (lamps, sconces, pendants). Do not introduce light from walls or areas without logical entry points. Maintain strong tonal separation between interior shadows and exterior highlights; window views should read clear, saturated, and contrast-rich, while the interior retains depth, drama, and editorial punch.

Generate a 50mm detail shot looking upward at the roofline with sharp architectural edges composed aesthetically against the sky. Sharp focus on roof edges with natural cloud formations in background. Frame the composition to emphasize the geometric shapes and lines of the roofline in a visually compelling arrangement. Preserve exact architecture, materials, and building details.

Transform this photo into a cinematic editorial image with harsh, directional shadows. Maintain the exact scene composition, original white balance, and ensure the overall exposure remains balanced — shadows should be dramatic but the image should not appear underexposed or muddy. Preserve all architectural and interior details. Light source should be the most logical and natural position.

Transform this photo into a cinematic editorial image with harsh, directional shadows. Correct any perspective distortion—ensure vertical lines are truly vertical and horizontal lines are level.

Maintain the exact scene composition after correction, preserve original white balance, and ensure the overall exposure remains balanced—shadows should be dramatic but the image should not appear underexposed or muddy. Preserve all architectural and interior details. The scene should feel well-lit yet moody. Derive all lighting direction from visible sources in the frame: windows, doors, architectural openings, and any practical fixtures (lamps, sconces, pendants). Do not introduce light from walls or areas without logical entry points.

Transform this photo into a cinematic editorial image with controlled, directional shadows. Correct any perspective distortion—ensure vertical lines are truly vertical and horizontal lines are level. Maintain the exact scene composition after correction, preserve original white balance, and ensure the overall exposure is bright and airy while retaining depth and dimension. Shadows should add shape and drama without making the space feel dark or dingy. Preserve all architectural and interior details with clarity. The scene should feel inviting, well-lit, and polished. Derive all lighting direction from visible sources in the frame: windows, doors, architectural openings, and any practical fixtures (lamps, sconces, pendants). Do not introduce light from walls or areas without logical entry points

Extreme close-up detail shot of harsh directional light casting crisp shadows across textured interior surface. Tight framing on shadow patterns and light interplay. Cinematic editorial style. Neutral white balance, balanced exposure — deep shadows without underexposure. Sharp texture detail in wood grain, fabric, or architectural elements. Preserve all architectural and interior details.

Extreme close-up detail shot with smooth tracking camera following harsh directional light as it grows and spreads across textured surface. Crisp shadow edges crawl and shift in real time. Camera moves with the light's path revealing texture in wood grain, fabric weave, architectural detail. Editorial film style. Neutral white balance, balanced exposure — deep dramatic shadows without underexposure. Shallow depth of field.

85mm close up detail shot of the main feature in the room. Cinematic editorial style. Neutral white balance, balanced exposure — deep shadows without underexposure. Sharp texture detail in wood grain, fabric, or architectural elements.

85mm close up detail shot of light being cast onto the bathroom cabinets. Cinematic editorial style. Neutral white balance, balanced exposure — deep shadows without underexposure. Sharp texture detail in wood grain, fabric, or architectural elements.

Cinematic detail shot of wood chair with light being cast - maintain white balance - soft bokeh - preserve all architectural and interior details.

Cinematic close up detail shot - maintain white balance - soft bokeh - preserve all architectural and interior details.